

Land Use / Land Cover (LULC) Classification

Experiment 1: Baseline Using Raw Multispectral Bands

Random Forest vs. XGBoost — Sentinel-2 Multi-Seasonal Band Analysis

Project Report

1. Objective

This experiment establishes a baseline for Land Use / Land Cover (LULC) classification using only raw Sentinel-2 multispectral bands acquired across four seasons. No spectral indices or additional feature engineering were applied. Two tree-based ensemble classifiers — Random Forest (RF) and XGBoost — were evaluated on the same dataset and feature configuration to enable a direct, controlled comparison.

2. Dataset and Feature Configuration

Each sample was represented by 40 raw spectral features, consisting of 10 Sentinel-2 bands from each of the four seasons. The configuration used for this experiment is summarized below.

Parameter	Configuration
Input Data	Sentinel-2 multispectral data
Temporal Coverage	4 Seasons (Q1, Q2, Q3, Q4)
Spectral Bands	B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12
Bands per Season	10
Total Features	40
Spectral Indices	None
Additional Feature Engineering	None
Number of LULC Classes	9
Samples per Class	3,000
Total Samples	27,000
Test Set	5,400 samples (20%)
Class Labels	10, 20, 30, 40, 50, 60, 80, 90, 95

3. Models Evaluated

Two ensemble tree-based models were trained using identical input features:

- Random Forest Classifier
- XGBoost Classifier

This setup allowed a direct comparison between the two models without introducing differences in the underlying feature space.

4. Overall Results

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Random Forest	85.00%	0.85	0.86	0.85	0.85

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
XGBoost	87.02%	0.87	0.87	0.87	0.87

Model Improvement

Relative to Random Forest, XGBoost achieved:

- +2.02 percentage points improvement in accuracy
- +0.02 improvement in Macro F1-score

5. Class-wise Performance

Random Forest

Class	Precision	Recall	F1-score
10	0.81	0.80	0.80
20	0.79	0.82	0.80
30	0.75	0.80	0.77
40	0.91	0.91	0.91
50	0.85	0.81	0.83
60	0.88	0.86	0.87
80	0.97	1.00	0.99
90	0.86	0.79	0.82
95	0.88	0.93	0.90

XGBoost

Class	Precision	Recall	F1-score
10	0.82	0.80	0.81
20	0.83	0.81	0.82
30	0.82	0.79	0.80
40	0.92	0.92	0.92
50	0.83	0.88	0.85
60	0.87	0.89	0.88
80	1.00	0.97	0.99
90	0.82	0.87	0.84

Class	Precision	Recall	F1-score
95	0.93	0.89	0.91

6. Key Observations

- Raw multispectral bands combined with multi-seasonal information provided a strong baseline for LULC classification, even without spectral indices.
- XGBoost outperformed Random Forest, achieving 87.02% accuracy compared with 85.00% for Random Forest.
- Macro F1-score increased from 0.85 (Random Forest) to 0.87 (XGBoost), indicating an improvement in overall class-wise performance.
- Class 80 was consistently the easiest class to classify, achieving an F1-score of 0.99 with both models.
- For Random Forest, Class 30 had the lowest F1-score (0.77); XGBoost improved this to 0.80.
- XGBoost showed particularly noticeable improvements for Classes 20, 30, 50, 60, and 90 compared with Random Forest.

7. Conclusion

Experiment 1 established the baseline performance of the LULC classification system using raw Sentinel-2 multispectral bands across four seasons. A total of 40 spectral features were used without incorporating spectral indices or additional feature engineering.

Among the two evaluated ensemble models, XGBoost achieved the best performance, with an accuracy of 87.02% and a macro F1-score of 0.87, outperforming Random Forest (85.00% accuracy, 0.85 macro F1-score).

These results demonstrate that raw multi-seasonal spectral information provides substantial discriminatory power for LULC classification, and establish a strong baseline against which the impact of subsequent feature engineering and experimental configurations can be evaluated.